Class 05

Clauses and Aliasingl

Constraints are rules applied on table columns to maintain accuracy, reliability, and integrity of the data.

Keys are used to identify each row in a table and create relationships between tables.

- **Types of constraints**
- **Types of keys**

Types of Constraints:

Constraint	Description	Example
NOT NULL	Ensures a column cannot have NULL values.	name VARCHAR(50) NOT NULL
UNIQUE	Ensures all values in a column are unique.	email VARCHAR(100UNIQUE
PRIMARY KEY	Uniquely identifies each record in a table.	id INT PRIMARY KEY
FOREIGN KEY	Links two tables together.	FOREIGN KEY (dept_id) REFERENCES department(id)
DEFAULT	Assigns a default value if none is provided.	status VARCHAR(10) DEFAULT 'active'
CHECK	Ensures that all values in a column meet a specific condition.	CHECK (age >= 18)

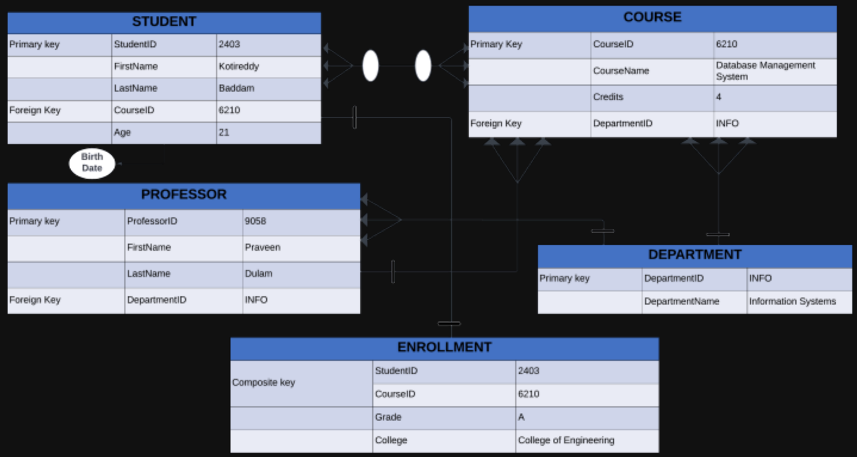
🗝️ Types of Keys:

Key Type	Description	Example
Primary Key	Uniquely identifies each record.	id INT PRIMARY KEY
Foreign Key	References a key in another table.	FOREIGN KEY (dept_id) REFERENCES department(id)
Candidate Key	All possible keys that can uniquely identify a row.	roll_no, email
Alternate Key	Candidate keys that are not selected as primary key.	If `roll_no` is PK, `email` is alternate key.
Composite Key	Combination of two or more columns to uniquely identify a row.	PRIMARY KEY (student_id, course_id)
Super Key	Any key that can uniquely identify a record (can include extra columns).	{roll_no, name}

Examples:

Create Table with Constraints:

```
CREATE TABLE students (
  student_id INT PRIMARY KEY,
  name VARCHAR(50) NOT NULL,
  age INT CHECK (age >= 18),
  email VARCHAR(100) UNIQUE,
  dept_id INT,
  FOREIGN KEY (dept_id) REFERENCES department(dept_id)
);
```



Altering a Table:

```
ALTER TABLE students
ADD COLUMN phone VARCHAR(15);
```

```
ALTER TABLE students
DROP COLUMN age;
```